

Data Workflow for Swiss Legal Entity Identifier (LEI) Data

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Overview

In this project I built a reproducible data workflow for the GLEIF LEI Golden Copy (Level 1, “Who is who”), filtered to Swiss entities (<https://www.gleif.org/en/lei-data/gleif-golden-copy/download-the-golden-copy>). The workflow covers ingestion, cleaning, exploratory analysis, three visualizations and a written summary, implemented in a Jupyter notebook (*data_workflow.ipynb*) with Python, NumPy, Pandas, Matplotlib and Seaborn. I picked this dataset because the later capstone projects will build on it, with entity matching for compliance checks as the long-term goal.

Dataset Description

The Legal Entity Identifier (LEI) is a 20-character code from the ISO 17442 standard that identifies legal entities which take part in financial transactions (Global Legal Entity Identifier Foundation [GLEIF], n.d.-e). GLEIF publishes the complete registry daily as the so-called Golden Copy (GLEIF, n.d.-a), and the data is free to use under a CC0 license (GLEIF, n.d.-c). The copy I downloaded on 6 August 2026 contained 3,395,297 records worldwide. Filtered to entities with a Swiss legal address, this gives the project dataset of 27,933 rows and 12 columns, far above the required minimum of 200 rows and 5 columns. The most important variables are the legal name, the legal form code (ISO 20275), the registration status, three registration dates and the address fields. Everything arrives as text, and the dates are ISO 8601 strings with mixed precision.

Workflow Description

Two small scripts prepare the data. *download_golden_copy.py* asks the GLEIF API for the newest file and downloads it, and *prepare_dataset.py* reads the raw 5 GB file in chunks and

keeps only the Swiss entities. The notebook loads the result with *pd.read_csv*. Two documented cleaning functions normalize the legal names (whitespace, quote characters, plus a lowercase column without legal form suffixes) and convert the date columns to UTC timestamps while counting every value that cannot be parsed. A reusable function (*explore_categorical*) reports missing values and the most frequent values of a column. I applied it to the legal form, the registration status and the city. Three labeled figures and a written summary close the workflow. All work is version-controlled in Git with a develop branch and one commit per work step.

Key Decisions and Assumptions

I filtered to Switzerland because it keeps the dataset manageable and matches the compliance topic of the later projects. All columns are loaded as strings and converted explicitly, so every type change is a visible step in the workflow. Writing the steps as small documented functions follows the goal of clean, reusable and reproducible analysis code (Danchev, 2022). For the dates I used *pd.to_datetime* with *errors="coerce"*, so invalid values become countable NaT values instead of stopping the run. The source CSV stays unchanged. In the working DataFrame the small whitespace and quote fixes go directly into *legal_name*, while the aggressive lowercase and suffix-stripped version is stored separately in *legal_name_normalized*. The three figures cover the composition (Figure 1), the growth over time (Figure 2) and the registration status per legal form (Figure 3). I also assume the field semantics documented in GLEIF's LEI-CDF standard (GLEIF, n.d.-d), especially that a lapsed status only means a missed renewal and not a closed company.

Results and Interpretation

Figure 1 shows a very concentrated picture. The code MVII (stock corporations, AG / SA) alone covers 15,183 entities or 54.4%, followed by 3EKS (GmbH / Sàrl) with 12.1%. The

reserved codes 8888 and 9999 together cover almost 10% of the records. In the current GLEIF code list, 8888 is used while a new code for a legal form is requested, and 9999 stands for entities without a separate legal form. The meaning of both codes changed with version 1.5 of the code list (GLEIF, n.d.-b).

Figure 1

Top 10 Legal Forms of Swiss LEI Entities

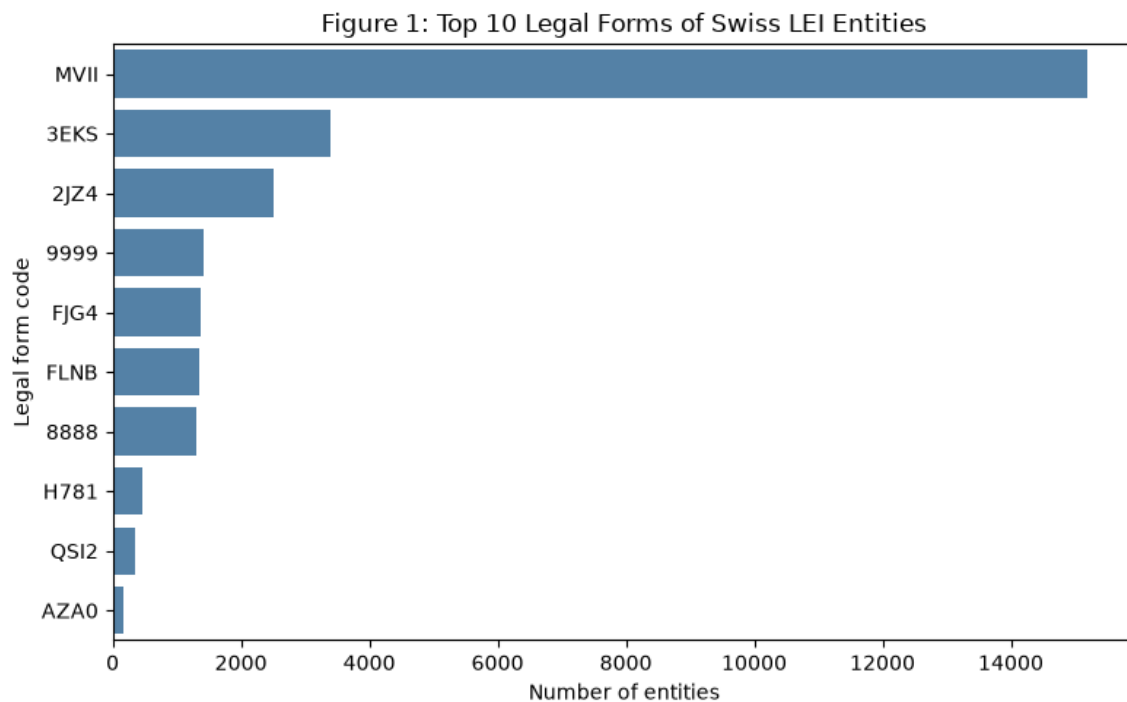


Figure 2 shows that almost 40% of all Swiss LEIs were registered in only two years, 6,089 in 2017 and 4,955 in 2018. The timing lines up with two events. The EU MiFID II transaction reporting rules, effective 3 January 2018, do not allow reportable trades for clients without an LEI (European Securities and Markets Authority [ESMA], 2017). And since 2017 the Swiss Federal Statistical Office has been the national LEI allocation office, known as “LEI-Suisse” (Federal Statistical Office [FSO], 2024). It issued 115 Swiss LEIs in 2016, 2,916 in 2017, and today it manages 41% of all Swiss LEIs. My analysis cannot prove what caused what, but demand and Swiss issuing capacity clearly grew at the same time. From 2019 on the curve

stays between roughly 1,400 and 2,300 registrations per year, and the low 2026 value only covers a part of the year.

Figure 2

New LEI Registrations in Switzerland per Year

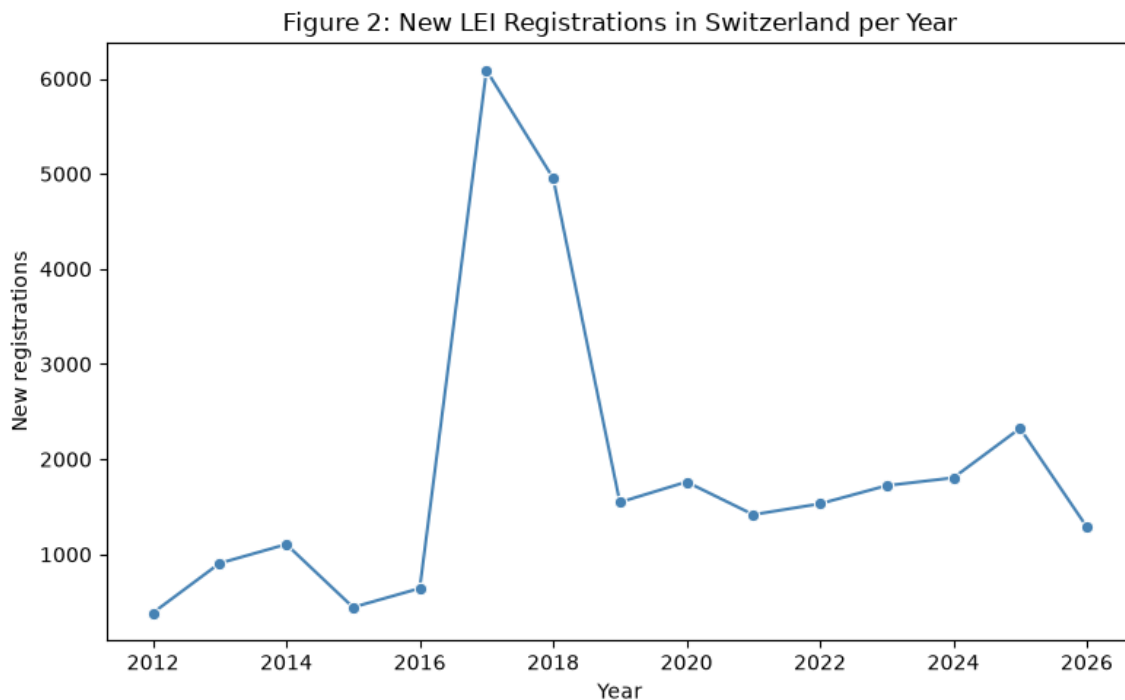
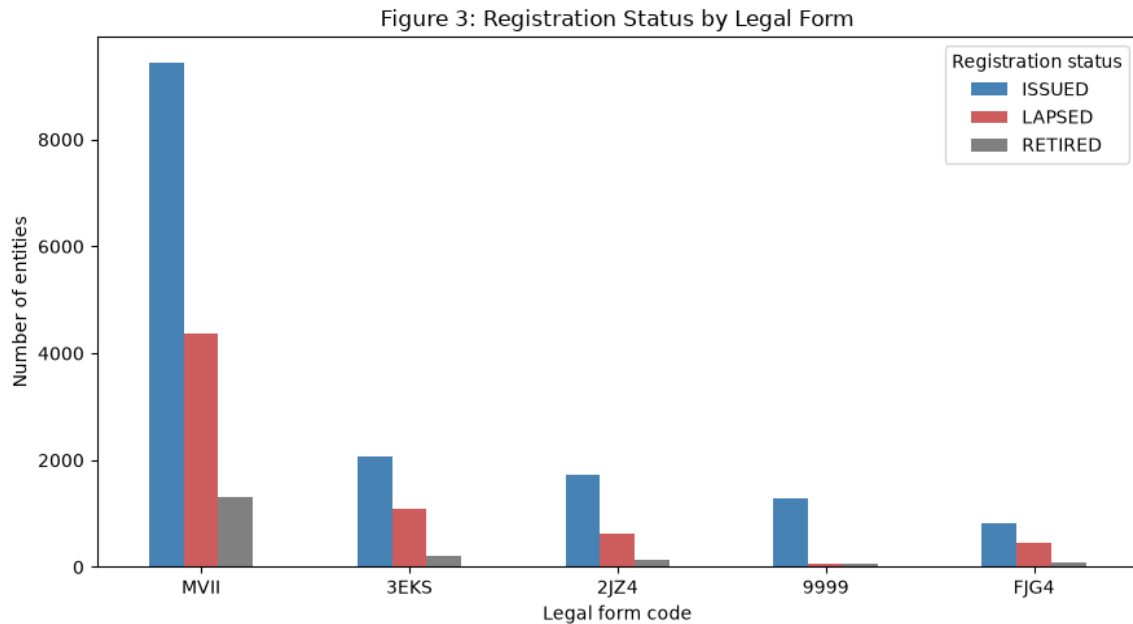


Figure 3 shows that the lapsed share differs clearly between the legal forms. It only shows the three main statuses, DUPLICATE and ANNULLED together are below 0.5%. Trusts (FJG4) and GmbH / Sàrl companies (3EKS) lapse at 33% and 32% of their registrations, stock corporations (MVII) at 29% and foundations (2JZ4) at 25%. The outlier is the code 9999 with only 5% lapsed. These are not newer registrations, 60% of them are from 2017 or earlier, and almost all of them are funds managed by one single LOU. Today 9999 stands for entities without a separate legal form (GLEIF, n.d.-b). When the current codes of these records were assigned is not visible in my dataset, it only contains the codes of the snapshot. My explanation for the low lapsed share is the professional fund administration behind these records. This is a hypothesis, not a result of my analysis. Overall 27.3% of all Swiss LEI registrations are currently lapsed.

Figure 3*Registration Status by Legal Form***Responsible Practice (Bias and Data Quality)**

Data quality is not only about accuracy, it is about fitness for use from the perspective of the people who work with the data (Wang & Strong, 1996), and this project has several risks of that kind. First, the dataset has a selection bias that no cleaning can fix. It only contains entities that needed an LEI, so it never represents all Swiss companies. Second, name normalization can merge different companies or split identical ones, and because umlauts and French or Italian accents are only common in some regions, a bad rule would distort exactly those regions. The notebook documents the scale of the problem, with eight raw spellings for Zürich (including one broken encoding) and over twenty strings for Geneva. A matching name alone also does not prove a duplicate, a stronger check would compare the legal form, the address and the register data as well, which is also how GLEIF's own duplicate check works (GLEIF, 2021). That duplicates are a real topic shows in the data itself, 100 records carry the official status DUPLICATE. Third, coercing unparseable dates can silently remove entities from time analyses,

which is why the cleaning function counts and reports every such value (zero in this run). Finally, reading “lapsed” as “company is gone” would be a serious misinterpretation. As mitigation, the source CSV stays untouched, the aggressive normalization is stored in its own column, every coercion is counted, and the limitations are written down in the notebook summary and the README.

Reproducibility

Reproducibility comes from scripted, documented and version-controlled steps instead of manual work (Wilson et al., 2017). The notebook combines code, documentation, figures and interpretation in one document, the literate programming style that Danchev (2022) describes as a base for reproducible research practice. I finalized it with “Restart Kernel and Run All”, so it runs top to bottom without errors and with sequential cell numbers. The dependencies are pinned in *requirements.txt*, created with *pip freeze* inside the project's own virtual environment.

Danchev (2022) uses the same mechanism to rebuild the original computing environment and to avoid errors caused by package updates. The dataset itself can be rebuilt from the official source with the two scripts, and the README explains all steps. The Git repository documents the progress with one commit per work step and a develop branch that is merged into main at the end, which follows the recommended version control practice for computational work (Wilson et al., 2017).

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